Exam 2024 solutions

B—bookwork, E—new example, T—new theory

1.

a) We denote

$$I_k = \begin{cases} 1, & \text{the k-th post is rotten} \\ 0, & \text{otherwise} \end{cases}$$ [3B]

$k+1, k+2, \ldots, k+7$ are 7 consecutive posts (if $k+i>17$, then $k+i \Leftarrow k+i-17$)

$R_k = \sum_{i=1}^{7} I_{k+i}$ is the number of rotten posts. [2B]

So, $E(R_k) = E(\sum_{i=1}^{7} I_{k+i}) = \sum_{i=1}^{7} E(I_{k+i}) = 7E(I_k) = \frac{56}{17} > 3$

where $E(I_k) = \frac{8}{17}$ (8 of 17 posts are rotten) [3B]

Since R_k should be integer, it must be the case that $P(R_k \geq 4) > 0$, for some k, i.e. there necessarily exists a run of 7 consecutive posts at least 4 of which are rotten. [2B]

The method given above is probabilistic of course, but marks will also be given if student uses a correct combinatorial method.

b)

i)

$$\mathbb{E}(X) = \sum_{m=0}^{\infty} m\mathbb{P}(X=m) = \sum_{m=0}^{\infty}\sum_{n=0}^{m-1} \mathbb{P}(X=m) = \sum_{n=0}^{\infty}\sum_{m=n+1}^{\infty} \mathbb{P}(X=m) = \sum_{n=0}^{\infty} \mathbb{P}(X>n).$$ [8T]

2 marks

each step

ii) Denote by r and b the numbers of red and blue balls, respectively. Let N be the number of balls drawn. From i), we have

$$\begin{aligned}\mathbb{E}(N) &= \sum_{n=0}^{r} \mathbb{P}(N > n) = \sum_{n=0}^{r} \mathbb{P}(\text{first } n \text{ balls are red}) \\ &= \sum_{n=0}^{r} \frac{r}{b+r} \frac{r-1}{b+r-1} \cdots \frac{r-n+1}{b+r-n+1} = \sum_{n=0}^{r} \frac{r!}{(b+r)!} \frac{(b+r-n)!}{(r-n)!} \qquad \text{[3T]} \\ &= \frac{r!\,b!}{(b+r)!} \sum_{n=0}^{r} \binom{n+b}{b} = \frac{b+r+1}{b+1}, \qquad \text{[2T]}\end{aligned}$$

Substituting in r =20 and b = 10, we obtain

$$E(N) = \frac{31}{11} \approx 3 \qquad \text{[2T]}$$

You may use another approach if you wish.

2.

a) The joint distribution of the samples is given by

$$f_X(x_1, \dots, x_n; \lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda x} = \lambda^n e^{-\lambda \sum x_i}$$

Then the log-likelihood function is [3E]

$$\ln f_X(x_1, \dots, x_n; \lambda) = n \ln \lambda - \lambda \sum x_i$$

Now take the derivatives:

$$\frac{\partial}{\partial \lambda} \ln f_X(x_1, \dots, x_n; \lambda) = \frac{n}{\lambda} - \sum x_i$$

$$\frac{\partial^2}{\partial \lambda^2} \ln f_X(x_1, \dots, x_n; \lambda) = -\frac{n}{\lambda^2} \quad [3E]$$

Then the Fisher information is given by

$$I = -E\left[-\frac{n}{\lambda^2}\right] = \frac{n}{\lambda^2} \quad [2E]$$

So the Cramer-Rao bound is

$$\mathrm{Var}\left[\hat{\lambda}\right] = \frac{1}{I(\lambda)} = \frac{\lambda^2}{n} = \frac{4}{20} = 0.2 \quad [2E]$$

b) Recall the Wiender-Hopf equation

$$\mathbf{c} = \mathbf{R}^{-1}\mathbf{r}$$

$$\sigma^2 = r_0 - \mathbf{r}^T \mathbf{R}^{-1} \mathbf{r} \quad [2E]$$

i) If *n*=1, the Wiener-Hopf equation trivially reads

$$R_Y(0) c_1 = R_Y(1)$$

Therefore,

$$c_1 = \frac{R_Y(1)}{R_Y(0)} = R_Y(1) \quad [2E]$$

because $R_Y(0) = 1$. Therefore,

$$c_1 = R_Y(1) = 0.1 \quad [2E]$$

$$\sigma^2 = 1 - r_1^2 = 1 - 0.1^2 = 0.99$$

ii) When n = 2, we have

$$\mathbf{R} = \begin{bmatrix} 1 & 0.1 \\ 0.1 & 1 \end{bmatrix}$$

$$\mathbf{r} = \begin{bmatrix} -0.2 \\ 0.1 \end{bmatrix} \quad \text{[3E]}$$

Thus the coefficient vector

[3E]

$$\mathbf{c} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \mathbf{R}^{-1}\mathbf{r} = \begin{bmatrix} 1 & 0.1 \\ 0.1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} -0.2 \\ 0.1 \end{bmatrix} = \begin{bmatrix} -0.212 \\ 0.121 \end{bmatrix}$$

Mean-square error

$$\sigma^2 = 1 - [-0.2 \quad 0.1] \begin{bmatrix} 1 & 0.1 \\ 0.1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} -0.2 \\ 0.1 \end{bmatrix} = 0.945 \quad \text{[3E]}$$

3.

a) Let

$$y = \cos(2\pi nu). \qquad (*)$$ [2E]

We have

$$\frac{dy}{du} = 2\pi n \cdot \sin(2\pi nu)$$ [2E]

Applying the formula of pdf, we obtain

$$f_Y(y) = \sum_{i=1}^{2n} \frac{1}{2\pi n \cdot |\sin(2\pi n u_i)|}$$ [3E]

because given *y*, there are 2*n* solutions u_i to Eq. (*) above.

Since $|\sin(2\pi n u_i)| = \sqrt{1-y^2}$, we have

$$f_Y(y) = \sum_{i=1}^{2n} \frac{1}{2\pi n \cdot \sqrt{1-y^2}} = \frac{1}{\pi\sqrt{1-y^2}}, \quad -1 \le y \le 1$$ [3E]

b)

i) Let state A, B and C indicate the player that is not playing in the given game. The transition matrix is given by [2E]

$$\begin{bmatrix} P_{A\to A} & P_{A\to B} & P_{A\to C} \\ P_{B\to A} & P_{B\to B} & P_{B\to C} \\ P_{C\to A} & P_{C\to B} & P_{C\to C} \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$ [3E]

where $P_{A\to B} = \frac{1}{2}$ means A is not playing in the first game and B is beaten in this game and will not play in the following game.

The process is Markov chain because the results of the subsequent game are independent.

ii) We are looking for the probability that after 4 steps the chain returns to a given initial state. [2E]

If the initial state is A, then there are 6 ways to return to A after 4 steps:

$$A \to B \to A \to B \to A$$ [1E]

$$A \to B \to A \to C \to A$$ [1E]

$$A \to B \to C \to B \to A$$ [1E]

$$A \to C \to A \to B \to A$$ [1E]

$$A \to C \to A \to C \to A$$ [1E]

$$A \to C \to B \to C \to A$$ [1E]

Each path has probability 1/16, so the total probability is 6/16 = 3/8. [2E]

4.

a)

i) Obviously,

$$X_n = X_{n-1} + B_n \times Bet$$

where $B_n = \pm 1$ with equal probabilities if the game is fair. Therefore,

$$E[X_n | X_{n-1}, \dots, X_1, X_0] = E[X_{n-1} + B_n \times Bet | X_{n-1}, \dots, X_1, X_0] = X_{n-1}$$ [3B]

satisfying the definition of a martingale.

ii) N is a random variable with distribution

$$P(N = n) = 2^{-n}, \quad n = 1,2,3, \dots$$ [3B]

iii) However, by this time, he will have lost an amount L with mean ∞:

$$E[L] = \sum_{i=1}^{\infty} 2^{-n}(1 + 2 + \cdots + 2^{n-2}) = \infty$$ [2B] [2B]

This means that the gambler must be prepared to lose a huge amount of money!

b)

i) Of the 2n steps, suppose the chain goes upward for i steps, leftward for j steps. It returns to the origin if and only it also goes downward for i steps, rightward for j steps. Here we must have $i + j = n$. [2E]

Therefore, using the multinomial coefficient

$$P\{X_{2n} = (0,0)\} = \left(\frac{1}{4}\right)^{2n} \sum_{i+j=n} \frac{(2n)!}{(i!\,j!)^2}$$ [3E]

ii) The above formula may be rewritten as

$$P\{X_{2n} = (0,0)\} = \left(\frac{1}{4}\right)^{2n} \binom{2n}{n} \sum_{i+j=n} \frac{(n)!}{(i!\,j!)} \frac{(n)!}{(i!\,j!)}$$

$$= \left(\frac{1}{4}\right)^{2n} \binom{2n}{n} \sum_{i+j=n} \binom{n}{i} \binom{n}{j}$$ [2T]

Using the identity

$$\sum_{i+j=n} \binom{n}{i} \binom{n}{j} = \binom{2n}{n}$$ [2T]

we obtain

$$P\{X_{2n} = (0,0)\} = \left(\frac{1}{4}\right)^{2n}\binom{2n}{n}^2 = \left(\frac{1}{4}\right)^{2n}\left(\frac{(2n)!}{n!\,n!}\right)^2 \quad [2T]$$

Using Stirling's formula, we obtain

$$P\{X_{2n} = (0,0)\} \approx \left(\frac{1}{4}\right)^{2n}\left(\frac{\sqrt{2\pi(2n)}\left(\frac{2n}{e}\right)^{2n}}{\left(\sqrt{2\pi n}\left(\frac{n}{e}\right)^{n}\right)^2}\right)^2 = \frac{1}{\pi n} \quad [2T]$$

Finally, we find that

$$\sum_n P\{X_{2n} = (0,0)\} \approx \frac{1}{\pi}\sum_{n\geq 1} n^{-1} = \infty \quad [2T]$$

because the sum diverges.

We therefore conclude that the origin is a recurrent state.